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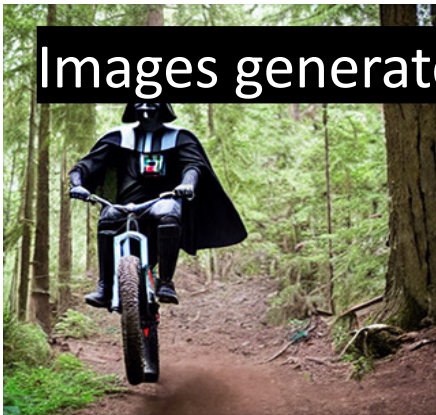
Bridging Visual Understanding and Generation with Multimodal LLMs

Xihui Liu

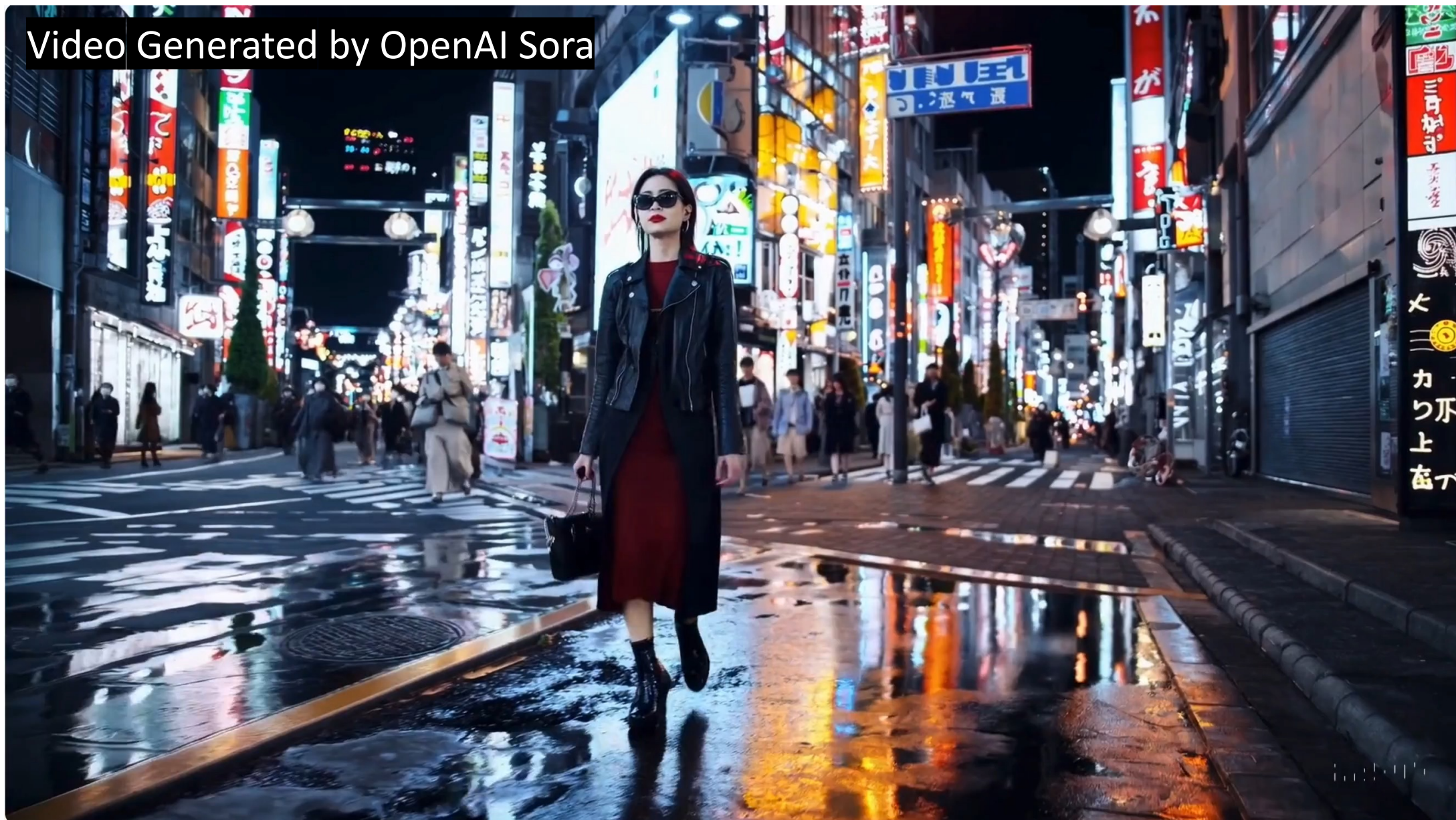
Presenter: Yukun Huang

The University of Hong Kong

Images generated by Stable Diffusion



Video Generated by OpenAI Sora



GPT-4Vision: Multimodal LLM

OpenAI INTRODUCES GPT-4V

Large Language Models



Vision Encoder

Text Prompt: Where is the tower building in the image?



The tower in the image is the **Space Needle**, an iconic observation tower in **Seattle**, Washington, USA. It is one of the **most recognizable landmarks** in the city and offers panoramic views of Seattle and the surrounding area.



Visual Understanding and Visual Generation

Visual Understanding

Open-world perception and understanding

Strong reasoning abilities

Commonsense knowledge

Visual Generation

Modeling data distribution

Generating photorealistic images and videos

User controls through language prompts

A collection of Multimodal GenAI Tools

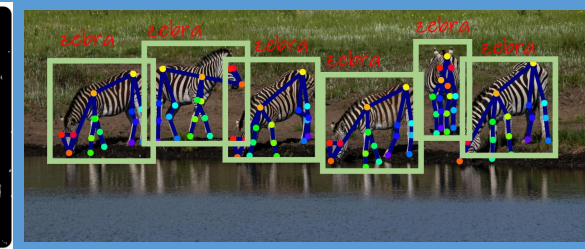
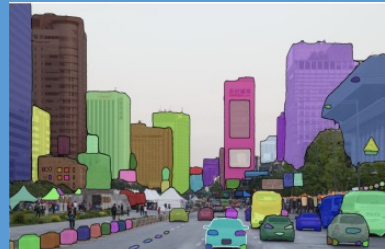
Foundation models for generation

(text-to-image, text-to-video, image-to-video, etc.)



Foundation models for perception

(detection, pose, segmentation, depth, edge, etc.)



Adapters and approaches built on foundation models

(controllable gen, image editing, personalized generation, image-to-image, etc.)



A detailed illustration of a futuristic robot artist in a grand, classical hall. The robot, with a white and gold body and large circular eyes, is painting a vibrant, colorful galaxy on an easel. It is surrounded by various art supplies like paint palettes, brushes, and jars. The hall features ornate columns, a chandelier, and a checkered floor. A large, colorful, multi-colored cable or pipe structure hangs from the ceiling, adding a futuristic touch to the scene.

Now we have so many AIGC Models
and Tools.

What is next?

Are they enough for users' real applications?

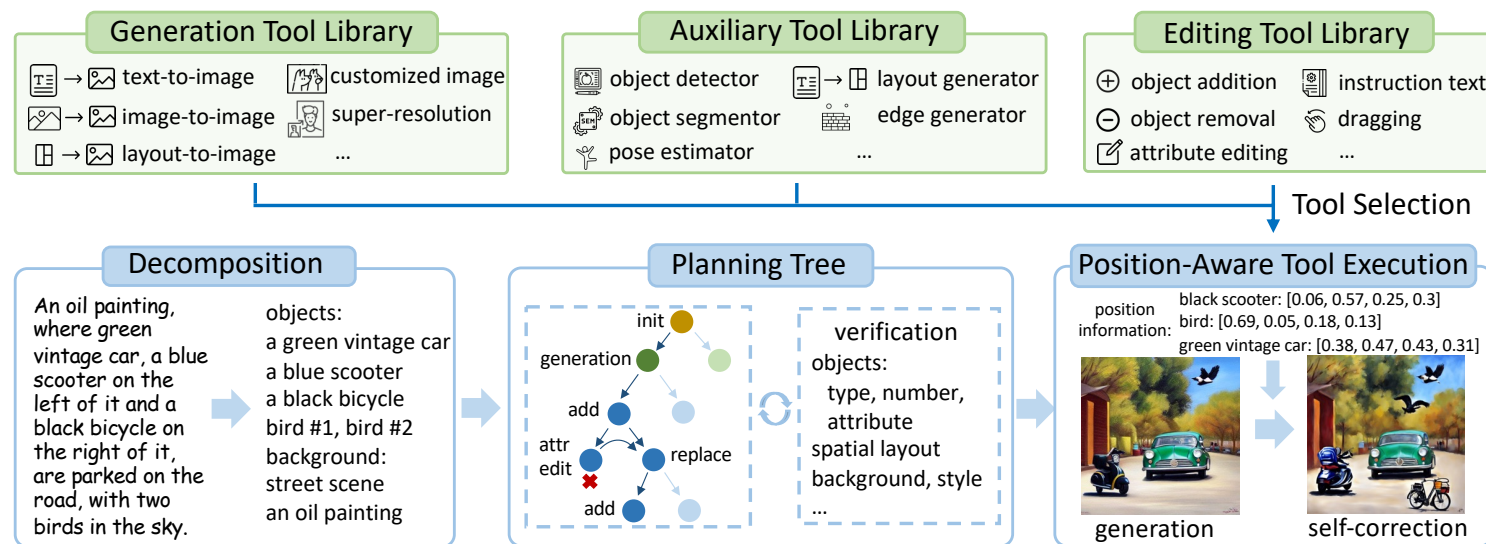
- Real applications are much more complex.
- They require combinations of different tools.
- Artists, designers, and users spend a lot of time to try and compose tools for specific applications.

Are they enough for users' real applications?

- Real applications are much more complex.
- They require combinations of different tools.
- Artists, designers, and users spend a lot of time to try and compose tools for specific applications.
- Can we automate this process?
- MLLM as agents!

GenArtist

- A unified system for image generation, editing, verification and self correction.
- LLM agent conducts problem decomposition, tool selection, and self-correction.



Wang et al., GenArtist: Multimodal LLM as an Agent for Unified Image Generation and Editing, NeurIPS 2024 (Spotlight).

GenArtist Results: Text-to-Image Generation

User Input:

An icy landscape. A vast expanse of snow-covered mountain peaks stretches endlessly. Beneath them is a dense forest and a colossal frozen lake. Three people are boating in three boats separately in the lake. Not far from the lake, a volcano threatens eruption, its rumblings felt even from afar. Above, a ferocious red dragon dominates the sky and commands the heavens, fueled by the volcano's relentless energy flow.



GenArtist output:



SDXL output:



Midjourney output:



DALL-E 3 output:



GenArtist Results: Image Editing

User Input:

In a vibrant summer village in the daylight, the snow has melted and lush green trees dot the landscape, while houses blend seamlessly. In the distance, some mountains retain a hint of snow, a small river is in the middle of the village.



GenArtist output:



GenArtist Results: Personalized Image Generation

User Input:

A winter beach at sunset, snow covered everywhere,
with a layout similar to the given beach.



GenArtist output:



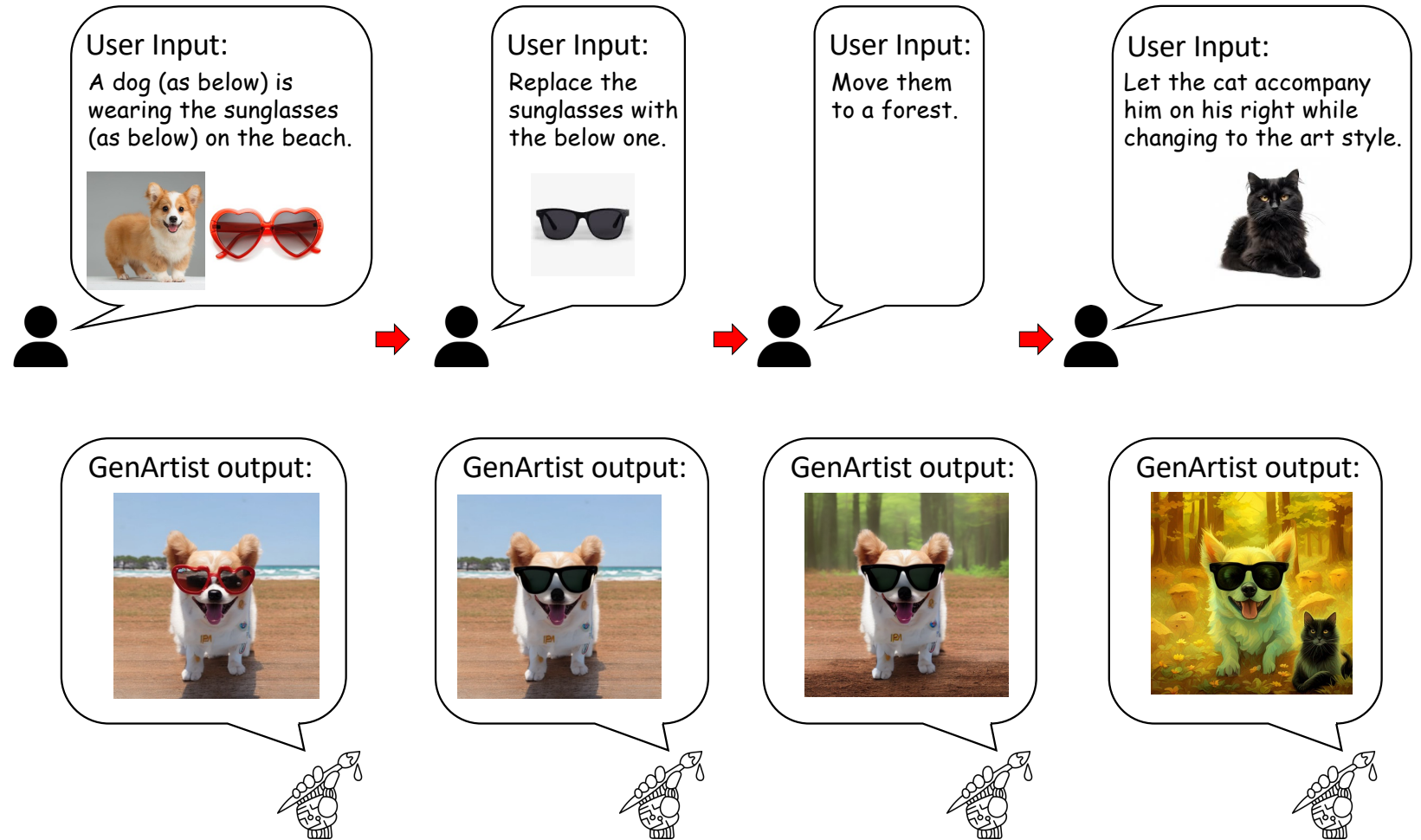
User Input:

a  with a 
and a  on it,
two  near it,
in front of the Egyptian Pyramid.

GenArtist output:

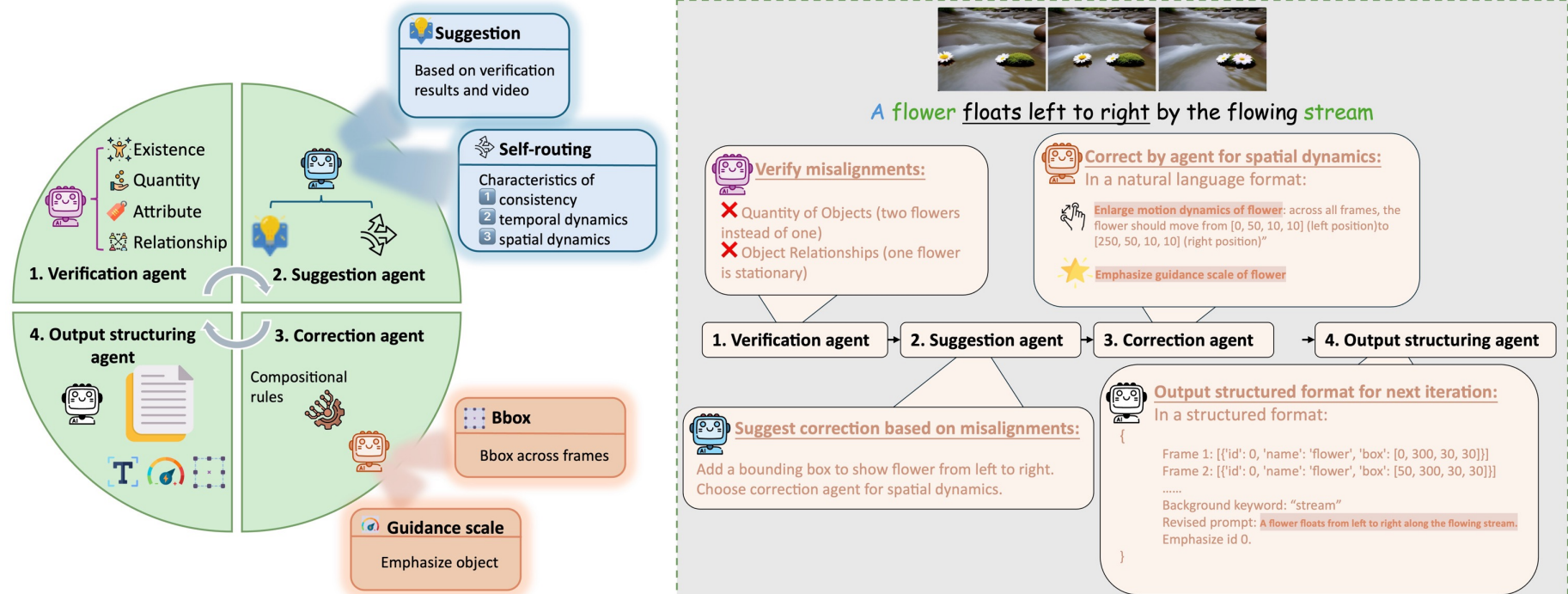


GenArtist
Results:
Multi-round
Interactive
Image
Generation



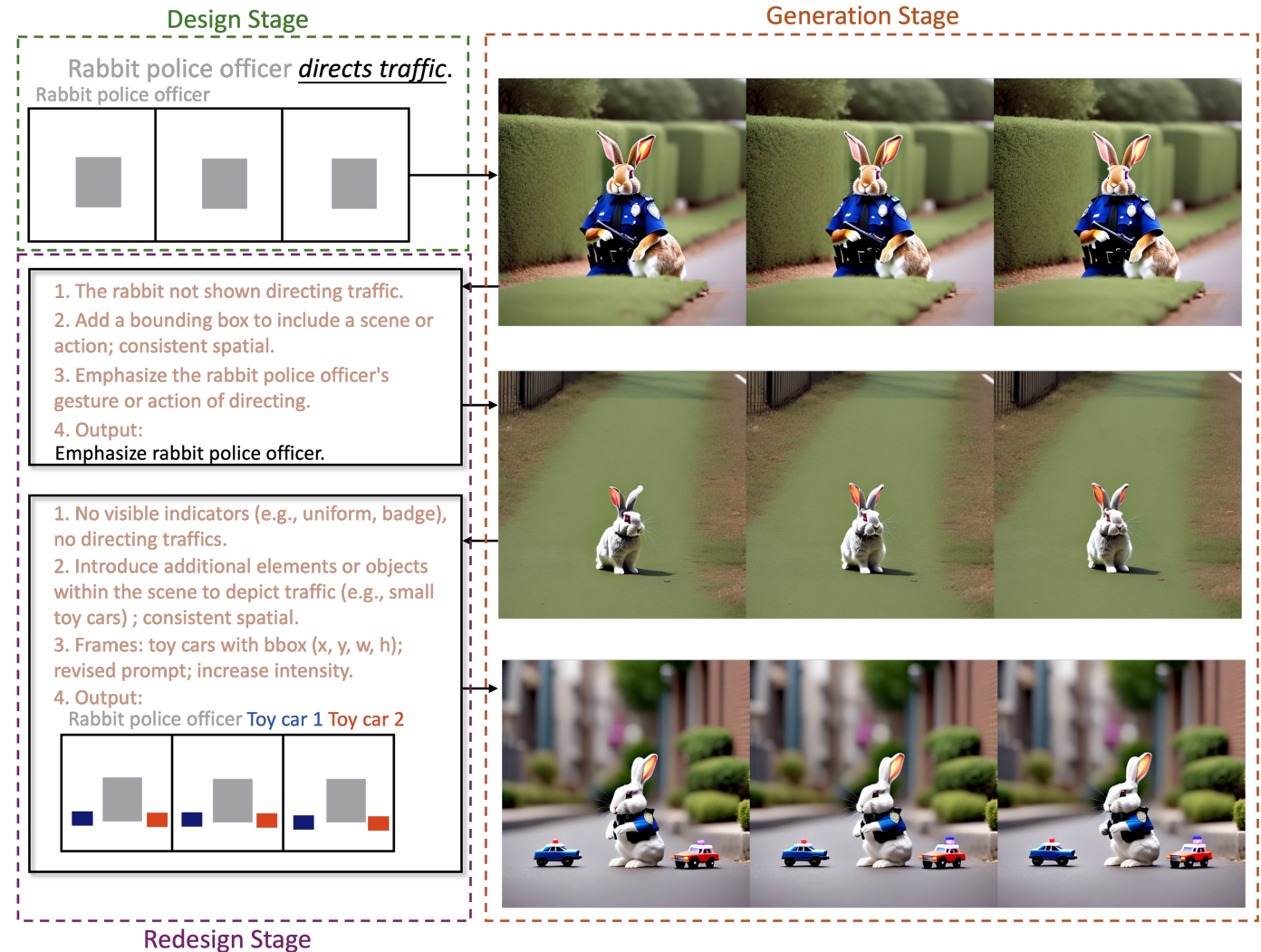
GenMAC

- Multi-agent iterative workflow improves compositional text-to-video generation.



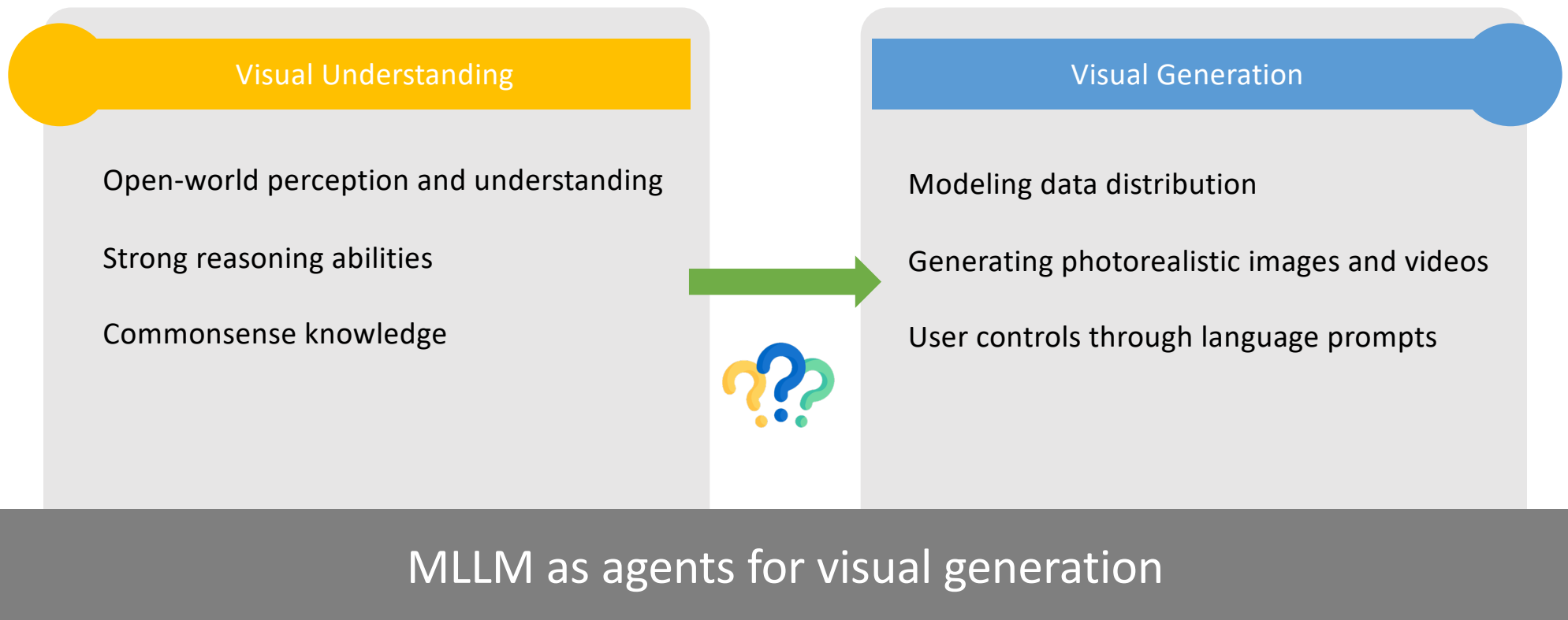
Huang et al., GenMAC: Compositional Text-to-Video Generation with Multi-Agent Collaboration, 2025.

GenMAC

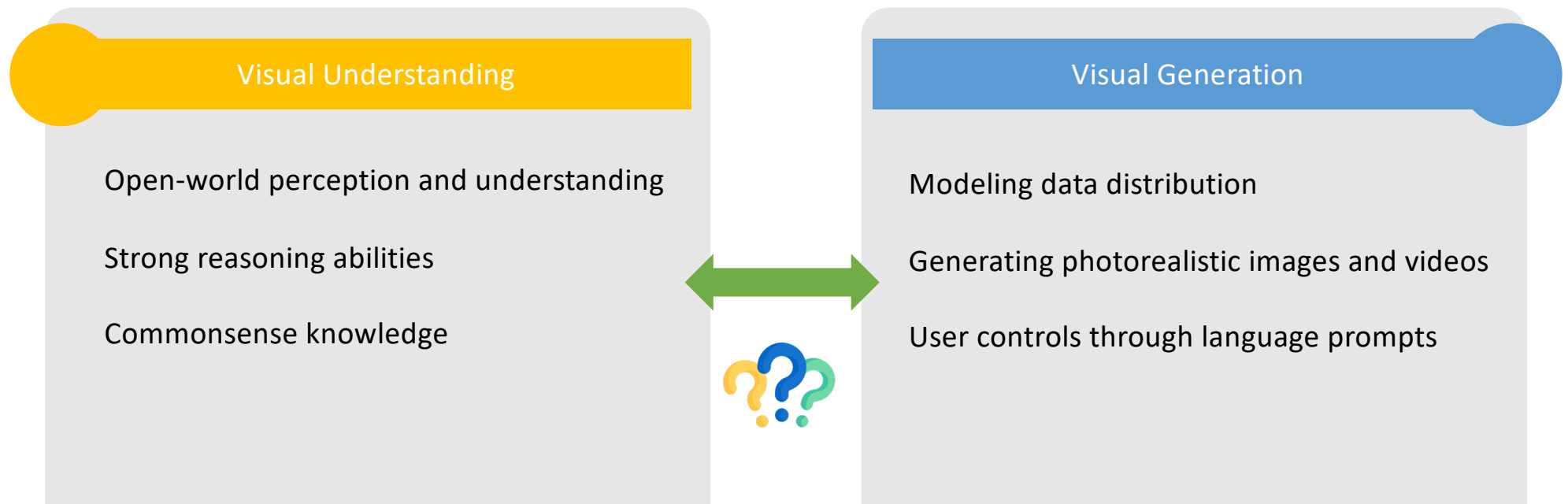


Huang et al., GenMAC: Compositional Text-to-Video Generation with Multi-Agent Collaboration, 2025.

Visual Understanding and Visual Generation

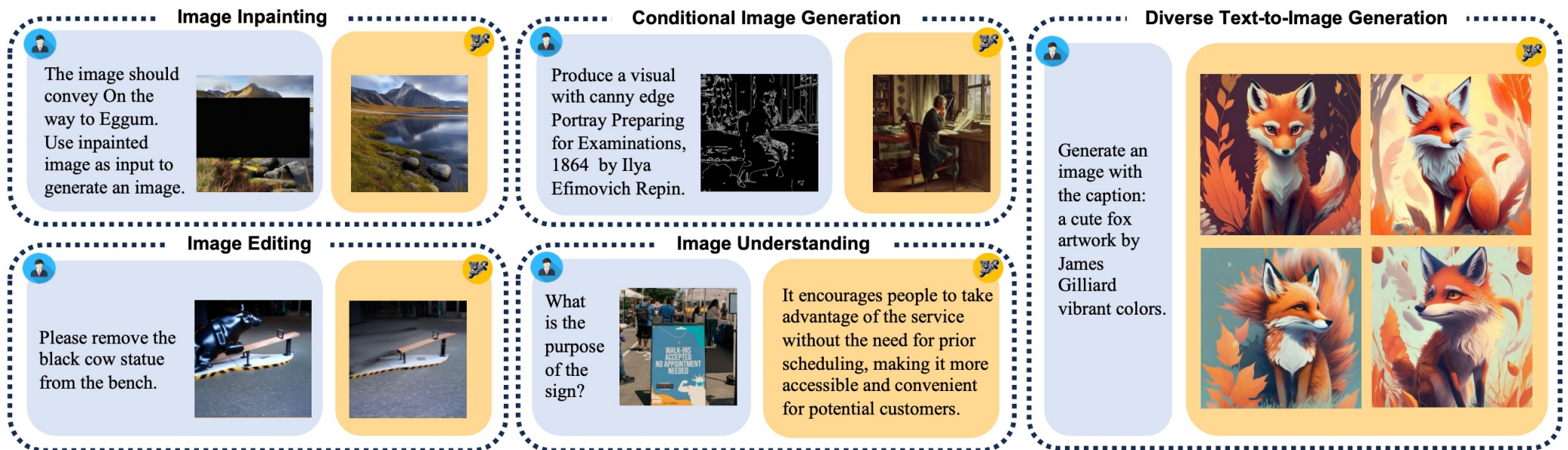


Visual Understanding and Visual Generation

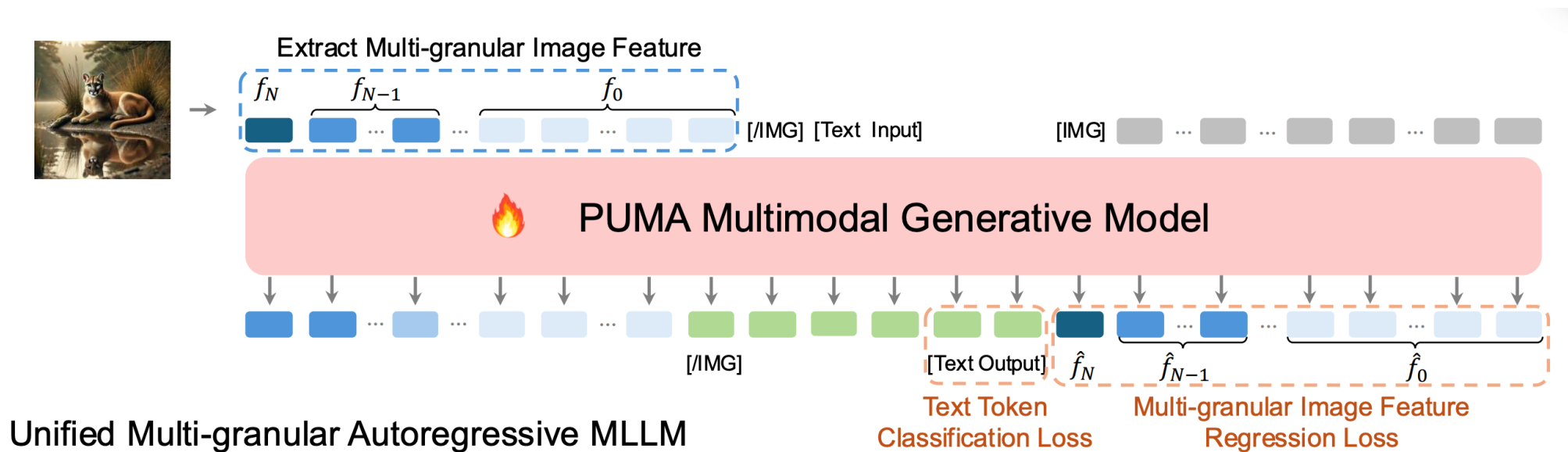


Can we have MLLM for unified visual understanding and generation?

Unified Visual Generation and Understanding

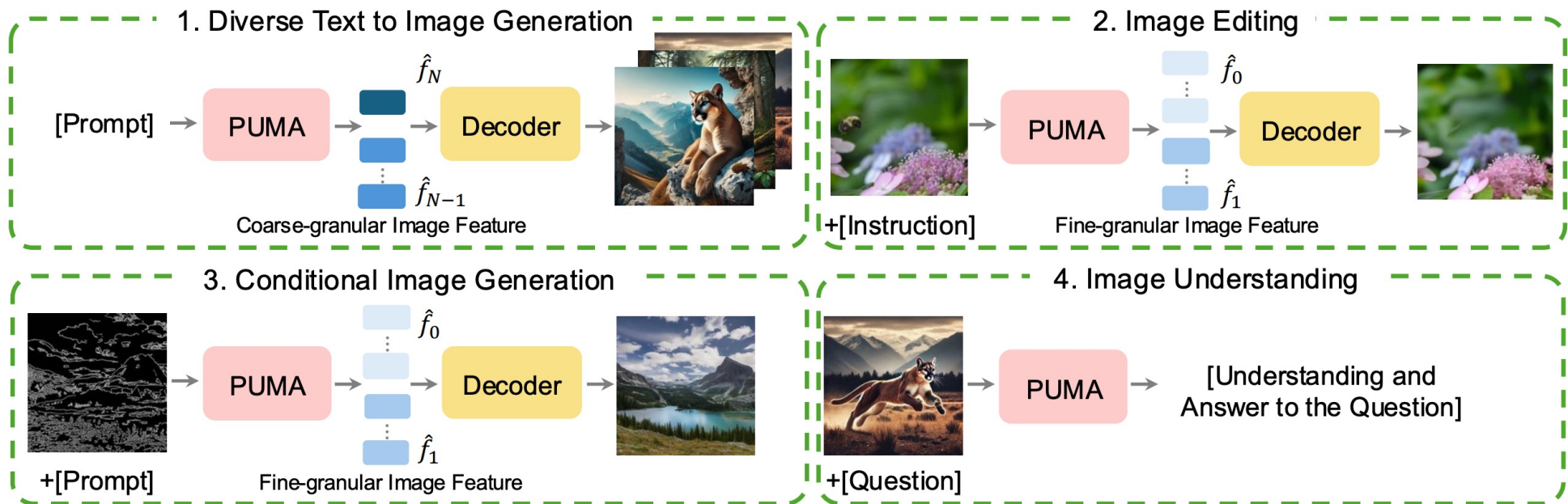


Unified Autoregressive Multi-granular MLLM

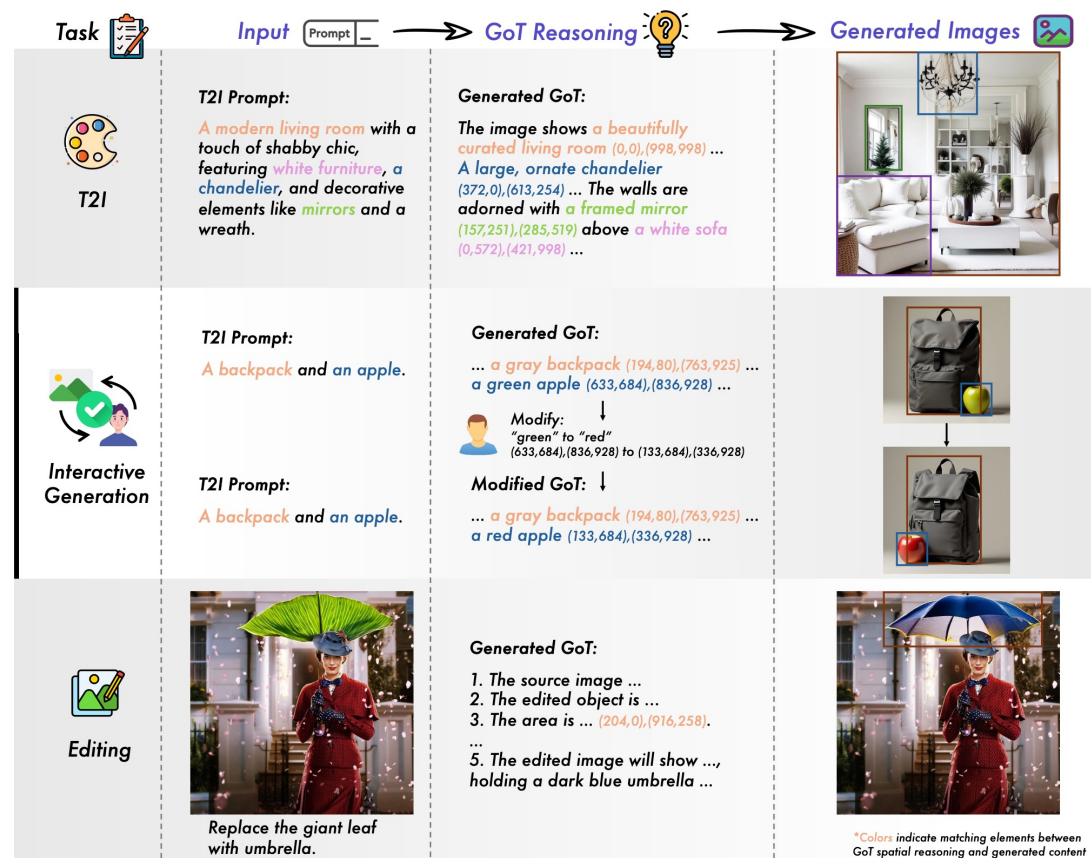


Fang et al., PUMA: Empowering Unified MLLM with Multi-Granular Visual Generation, 2024.

Unified Modeling of Downstream Tasks

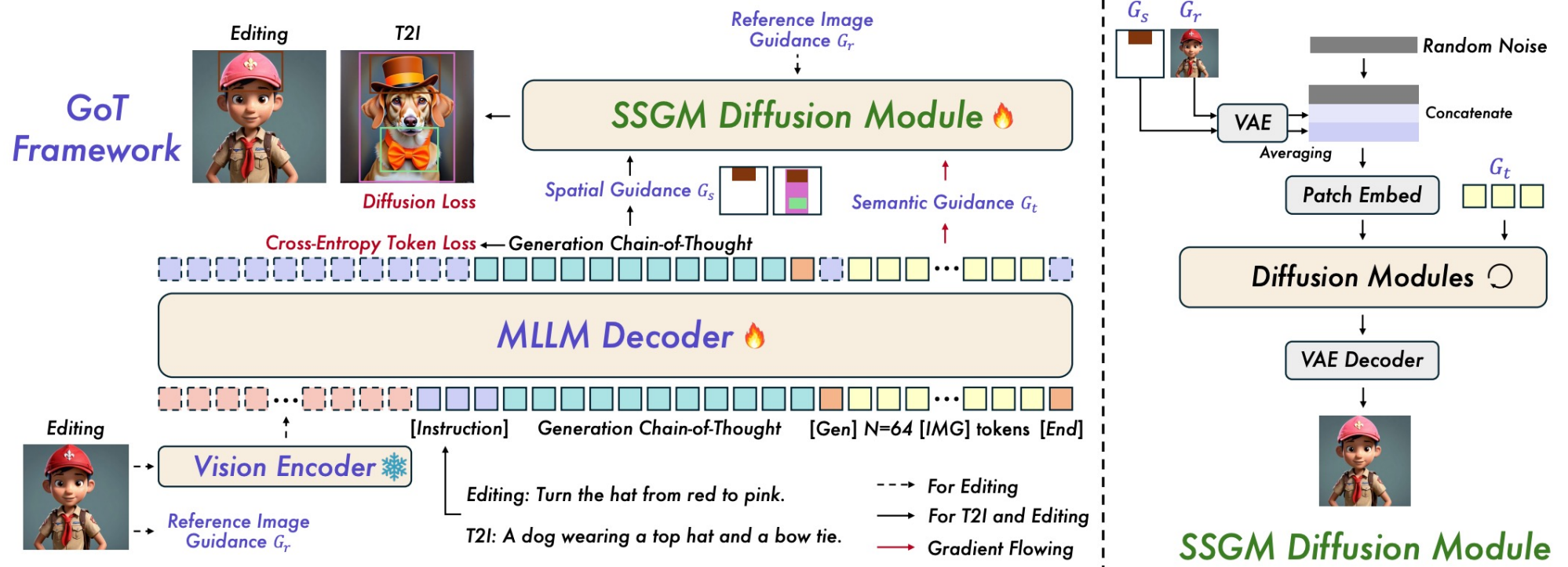


Chain-of-Thought Reasoning for Visual Generation



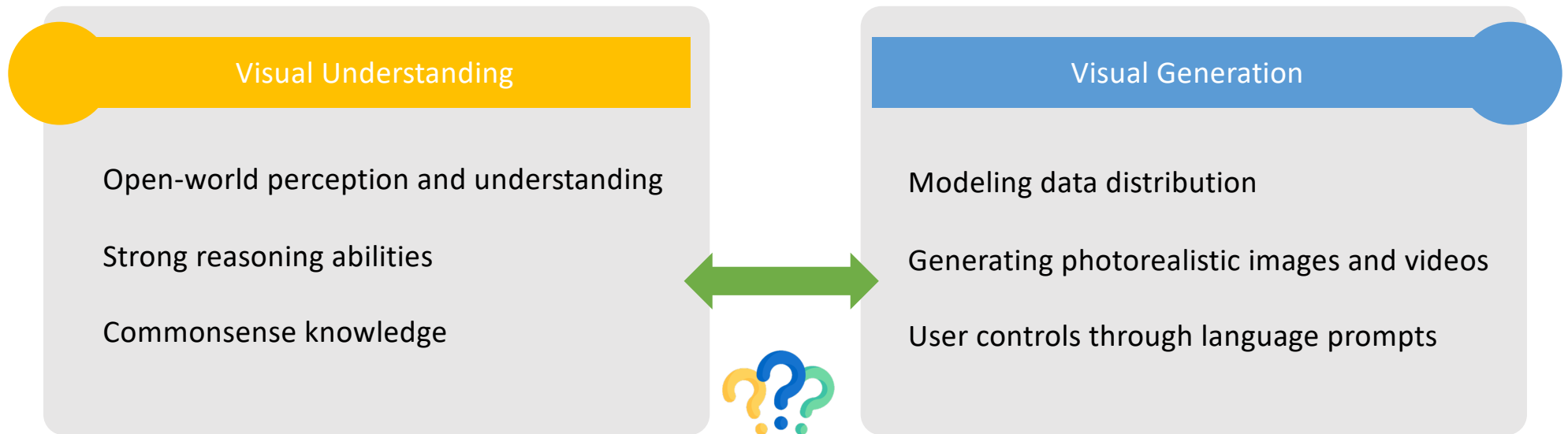
Fang et al., GoT: Unleashing Reasoning Capability of Multimodal Large Language Model for Visual Generation and Editing, 2025.

Chain-of-Thought Reasoning for Visual Generation



Fang et al., GoT: Unleashing Reasoning Capability of Multimodal Large Language Model for Visual Generation and Editing, 2025.

Thank you!



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